A Rigorous Assessment of a Proposed Label-Setting Certificate for Stochastic Shortest-Path Games

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Abstract

We analyze the manuscript *A Label-Setting Certificate for Two-Player Stochastic Shortest Path Games*. Its central label-setting theorem is mathematically valid after one compressed comparison step is supplied. Thus, there is no fatal error in the core correctness argument. However, several subsidiary claims are false or materially overstated. More importantly, the certificate does not solve the open direction posed by Gaspard and Vladimirsky concerning useful a priori label-setting criteria for general stochastic shortest-path games. The support-separation hypothesis already gives an explicit topological order after the certified action pruning, so the reduced problem is triangular and can be solved by a direct sweep without Dijkstra's method. We further prove that, in the manuscript's finite pure-action model, existence of the proposed certificate is equivalent to existence of the desired strictly value-decreasing optimal selector; in particular, the certificate may be obtained by setting its lower and upper bounds equal to the unknown value function itself. Consequently, the result is best classified as weak partial progress for a strongly restricted, reducibly acyclic subclass, not as a solution of the stated open problem.

1 Verdict and relation to the open problem

The source paper asks for criteria under which label-setting methods apply to stochastic shortest-path games whose transition law depends on the actions of two antagonistic players [2]. The associated problem entry asks, more specifically, for verifiable sufficient conditions stated a priori in terms of local game data [3]. The manuscript under review proposes a certificate consisting of value bounds, a set of "causal" minimizing actions, a robust dominance condition for all other minimizing actions, and a strict support-separation condition.

Assessment. The main correctness theorem has no fatal mathematical error, although its proof omits one nontrivial comparison that must be inserted. Several surrounding statements are false as written, including the assertion that certified noncausal actions are never optimal and the assertion that the transience constraints are jointly linear in their unknown weight and contraction factor. The manuscript does *not* solve the open problem. Its valid content is at most weak partial progress: after pruning, its hypotheses already make the system explicitly triangular in a known order, and existence of the certificate is equivalent to the implicit monotone-causality property that one was trying to certify.

The purpose of this note is fourfold. First, we give a complete proof of the valid core theorem. Second, we show that the support-separation condition itself yields an explicit acyclic ranking, making the proposed Dijkstra mechanism unnecessary. Third, we prove an equivalence theorem showing that the certificate can encode the unknown solution exactly. Finally, we record several scope and correctness limitations.

2 Model and proposed certificate

Let

$$X = N \cup \{t\}, \qquad N = \{1, \dots, n\},$$

where t is an absorbing target. At $x \in N$, Min chooses $a \in A_x$ and Max chooses $b \in B_x$, where all action sets are finite. The one-stage cost is $c_x^{ab} \geq 0$, and p_{xy}^{ab} is the probability of transitioning to $y \in X$. We set $v_t = 0$ and write

$$Q_x^{ab}(v) := c_x^{ab} + \sum_{y \in N} p_{xy}^{ab} v_y, \qquad v \in \mathbb{R}^N. \tag{1}$$

The manuscript studies the pure-action operator

$$(Tv)_x = \min_{a \in A_x} \max_{b \in B_x} Q_x^{ab}(v). \tag{2}$$

As discussed in section 6, this is the appropriate operator for a sequential move in which Max responds after observing Min's action, or for a simultaneous game possessing the relevant pure saddle structure. It is not the general Shapley operator for a finite simultaneous zero-sum game.

The manuscript imposes the following strong uniform transience assumption: there exist $w \in \mathbb{R}^N_{>0}$ and $\rho < 1$ such that

$$\sum_{y \in N} p_{xy}^{ab} w_y \leq \rho w_x \qquad \text{for every } x \in N,\ a \in A_x,\ b \in B_x. \tag{3}$$

Define $\|v\|_w = \max_{x \in N} |v_x|/w_x$.

Lemma 2.1 (Contraction and uniqueness)**.** *Under (3), T is monotone and is a contraction with modulus at most ρ in $\|\cdot\|_w$. Hence T has a unique fixed point $U \in \mathbb{R}^N$.*

Proof. For every x, a, b and $v, z \in \mathbb{R}^N$,

$$\begin{aligned} |Q_x^{ab}(v) - Q_x^{ab}(z)| &\leq \sum_{y \in N} p_{xy}^{ab} |v_y - z_y| \\ &\leq \|v - z\|_w \sum_{y \in N} p_{xy}^{ab} w_y \leq \rho w_x \|v - z\|_w. \end{aligned}$$

Taking a maximum or a minimum over a finite family cannot increase a uniform pointwise Lipschitz bound. Thus $|(Tv)_x - (Tz)_x| \leq \rho w_x \|v - z\|_w$ for every x, proving contraction. Monotonicity follows from the nonnegativity of the transition probabilities. The fixed-point conclusion is Banach's theorem. □

A proposed certificate consists of vectors $\ell, u \in \mathbb{R}^N_+$, a number $\delta > 0$, and nonempty sets $A_x^{\mathrm{c}} \subseteq A_x$. Set $\ell_t = u_t = 0$. The four conditions are:

(C1) Value box.

$$\ell \leq T\ell, \qquad Tu \leq u, \qquad \ell \leq u. \tag{4}$$

(C2) Support separation.
For every $x \in N$, $\alpha \in A_x^{\mathrm{c}}$, and $b \in B_x$,

$$p_{xx}^{\alpha b} < 1, \tag{5}$$

and, for every $y \in X \setminus \{x\}$,

$$p_{xy}^{\alpha b} > 0 \quad \Longrightarrow \quad u_y + \delta \leq \ell_x. \tag{6}$$

(C3) Robust pruning.

For every $a \in A_x \setminus A_x^{\mathrm{c}}$, there is an $\alpha = \alpha(a) \in A_x^{\mathrm{c}}$ such that, for every $b' \in B_x$, there is a $b = b(a, b') \in B_x$ satisfying

$$Q_x^{\alpha b'}(v) \le Q_x^{ab}(v) \qquad \text{for every } v \in [\ell, u]. \tag{7}$$

(C4) Nonemptiness.

$A_x^{\mathrm{c}} \neq \varnothing$ for every $x \in N$.

Here $[\ell, u] = \{v \in \mathbb{R}^N : \ell \le v \le u\}$. Since the difference in (7) is affine in v, it is enough to check one endpoint choice per coordinate: if $d_y = p_{xy}^{\alpha b'} - p_{xy}^{ab}$, then (7) is equivalent to

$$c_x^{\alpha b'} - c_x^{ab} + \sum_{y:d_y \ge 0} d_y u_y + \sum_{y:d_y<0} d_y \ell_y \le 0. \tag{8}$$

3 The valid core theorem

We first isolate the three elementary ingredients needed for the correctness proof.

Lemma 3.1 (Certified bounds)**.** *If* (4) *holds, then* $\ell \le U \le u$.

Proof. By monotonicity, $\ell \le T\ell \le T^2\ell \le \cdots$. Contraction implies $T^k\ell \to U$, so $\ell \le U$. Similarly, $u \ge Tu \ge T^2u \ge \cdots \to U$. □

Lemma 3.2 (Correct meaning of robust pruning)**.** *Assume* (C3). *For every* $v \in [\ell, u]$ *and every* $x \in N$,

$$(Tv)_x = \min_{\alpha \in A_x^{\mathrm{c}}} \max_{b \in B_x} Q_x^{\alpha b}(v). \tag{9}$$

Thus every action outside A_x^{c} *may be removed without changing the operator on the certified box.*

Proof. Fix an excluded action a and its corresponding α. For every b', condition (7) gives a b with

$$Q_x^{\alpha b'}(v) \le Q_x^{ab}(v) \le \max_{\bar{b} \in B_x} Q_x^{a\bar{b}}(v).$$

Taking the maximum over b' yields

$$\max_{b'} Q_x^{\alpha b'}(v) \le \max_{b} Q_x^{ab}(v).$$

Hence no excluded action produces a strictly smaller worst-case branch value than all causal actions. Minimizing over all actions is therefore the same as minimizing over A_x^{c}. □

The next lemma eliminates a self-loop after all nonself successor values are known.

Lemma 3.3 (Scalar self-loop elimination)**.** *Let* Γ *and* B *be finite. For* $\alpha \in \Gamma$ *and* $b \in B$*, let* $r_{\alpha b} \in \mathbb{R}$ *and* $q_{\alpha b} \in [0, 1)$*. Define*

$$F(z) = \min_{\alpha \in \Gamma} \max_{b \in B} \bigl(r_{\alpha b} + q_{\alpha b} z\bigr).$$

Then F *has the unique fixed point*

$$z^* = \min_{\alpha \in \Gamma} \max_{b \in B} \frac{r_{\alpha b}}{1 - q_{\alpha b}}. \tag{10}$$

Proof. For each α, set

$$g_\alpha(z) = \max_b (r_{\alpha b} + q_{\alpha b} z), \qquad \theta_\alpha = \max_b \frac{r_{\alpha b}}{1 - q_{\alpha b}}.$$

For each b, $r_{\alpha b} + q_{\alpha b} z \le z$ if and only if $z \ge r_{\alpha b}/(1 - q_{\alpha b})$. Consequently,

$$g_\alpha(z) \le z \quad \Longleftrightarrow \quad z \ge \theta_\alpha. \tag{11}$$

Let $\theta = \min_\alpha \theta_\alpha$. If $z < \theta$, then $g_\alpha(z) > z$ for every α, so $F(z) > z$. Choose α^* with $\theta_{\alpha^*} = \theta$. At $z = \theta$, (11) gives $g_{\alpha^*}(\theta) = \theta$, while $g_\alpha(\theta) \ge \theta$ for every α; hence $F(\theta) = \theta$. If $z > \theta$, then $g_{\alpha^*}(z) < z$, so $F(z) < z$. Thus θ is the unique fixed point, and it equals the right-hand side of (10). $\square$

We now state the label-setting construction. Let $S \subseteq X$ contain t, and suppose exact values $\widehat{U}_y = U_y$ have already been assigned for $y \in S$. A causal action $\alpha \in A_x^{\mathrm{c}}$ is S-admissible at $x \notin S$ if every nonself successor under every Max response is already in S:

$$p_{xy}^{\alpha b} > 0,\ y \ne x \quad \Longrightarrow \quad y \in S \qquad \text{for every } b \in B_x. \tag{12}$$

For such an action, define

$$\Theta_S(x, \alpha) := \max_{b \in B_x} \frac{c_x^{\alpha b} + \sum_{y \in S} p_{xy}^{\alpha b} \widehat{U}_y}{1 - p_{xx}^{\alpha b}}, \tag{13}$$

and set

$$D_S(x) := \min_{\substack{\alpha \in A_x^{\mathrm{c}} \\ \alpha \text{ is } S\text{-admissible}}} \Theta_S(x, \alpha), \tag{14}$$

with $D_S(x) = +\infty$ if there is no admissible action. Starting from $S = \{t\}$ and $\widehat{U}_t = 0$, the manuscript repeatedly selects a state minimizing $D_S(x)$, assigns it that value, and inserts it into S.

Theorem 3.4 (Correctness of the certificate algorithm)**.** *Assume* (3) *and* (C1)–(C4). *Then:*

(i) $U \in [\ell, u]$ *and* U *satisfies the restricted equation*

$$U_x = \min_{\alpha \in A_x^{\mathrm{c}}} \max_{b \in B_x} Q_x^{\alpha b}(U). \tag{15}$$

(ii) *For every* x*, there is an optimal action* $\alpha_x \in A_x^{\mathrm{c}}$ *such that*

$$p_{xy}^{\alpha_x b} > 0,\ y \ne x \quad \Longrightarrow \quad U_y + \delta \le U_x \qquad \textit{for every } b \in B_x. \tag{16}$$

(iii) *The label-setting algorithm based on* (12)–(14) *returns* U *exactly after* n *state finalizations.*

Proof. By theorem 3.1, $U \in [\ell, u]$. Applying theorem 3.2 at $v = U$ gives (15). Choose an action α_x attaining the minimum in (15). If $p_{xy}^{\alpha_x b} > 0$ and $y \ne x$, then *(C2)* and the certified bounds imply

$$U_y \le u_y \le \ell_x - \delta \le U_x - \delta,$$

which proves (16).

It remains to prove the algorithm. We use induction on $|S|$. The base case $S = \{t\}$ is correct. Suppose all labels in the current S are exact.

First fix $x \notin S$ and an S-admissible action α. Put

$$r_{\alpha b} := c_x^{\alpha b} + \sum_{y \in S} p_{xy}^{\alpha b} U_y, \qquad q_{\alpha b} := p_{xx}^{\alpha b}.$$

Admissibility guarantees that these terms account for every successor other than x. From (15),

$$U_x \le \max_b \bigl(r_{\alpha b} + q_{\alpha b} U_x\bigr). \tag{17}$$

The sign characterization (11) now implies $U_x \le \Theta_S(x,\alpha)$. Minimizing over the admissible actions gives

$$U_x \le D_S(x) \qquad \text{for every } x \notin S. \tag{18}$$

This is the comparison omitted in the manuscript's proof: it does not follow merely by saying that D_S minimizes over fewer actions; one must use (17) and theorem 3.3.

Let x_0 minimize the true value over $X \setminus S$. For every causal action $\alpha \in A^{\mathrm{c}}_{x_0}$, every b, and every nonself successor y, condition *(C2)* gives

$$U_y \le u_y \le \ell_{x_0} - \delta \le U_{x_0} - \delta < U_{x_0}.$$

Such a y cannot remain outside S, by the minimality of U_{x_0} among unfinalized states. Hence every causal action at x_0 is S-admissible. Applying theorem 3.3 to the full restricted equation (15) at x_0 yields

$$D_S(x_0) = U_{x_0}. \tag{19}$$

Let x_* be selected by the algorithm. Then $D_S(x_*) \le D_S(x_0) = U_{x_0}$. On the other hand, minimality of x_0 and (18) give

$$U_{x_0} \le U_{x_*} \le D_S(x_*) \le U_{x_0}.$$

All inequalities are equalities. Therefore the new label is exact, closing the induction. Since one new state is finalized at every iteration, the procedure terminates after n finalizations. □

Remark 3.5. *The theorem proves only that excluded actions can be deleted without changing the value. It does not prove that they are never optimal; ties can occur, as shown in theorem 6.1.*

4 The support condition already makes the problem triangular

The main structural limitation is immediate from *(C2)*. Define the reduced nonself dependency graph

$$G_{\mathrm{c}} = (N, E_{\mathrm{c}}),$$

where $(x,y) \in E_{\mathrm{c}}$ if $x \ne y$ and there are $\alpha \in A^{\mathrm{c}}_x$ and $b \in B_x$ with $p^{\alpha b}_{xy} > 0$.

Proposition 4.1 (An explicit ranking is already supplied)**.** *Under* (C2)*, every edge* $(x,y) \in E_{\mathrm{c}}$ *satisfies*

$$\ell_y + \delta \le \ell_x. \tag{20}$$

Consequently, G_{c} is acyclic, and sorting states by nondecreasing ℓ_x gives a topological order of the reduced game.

Proof. If $(x,y) \in E_{\mathrm{c}}$, then *(C2)* gives $u_y + \delta \le \ell_x$. Since $\ell_y \le u_y$, (20) follows. Along every directed edge, ℓ decreases by at least δ in the forward transition direction. A directed cycle would therefore imply $\ell_x \le \ell_x - k\delta$ for some positive integer k, which is impossible. □

The self-loop at a state is local and can be eliminated by theorem 3.3. Thus no Dijkstra search is needed.

Corollary 4.2 (Direct topological solution)**.** *Let $x_1, \dots, x_n$ be ordered so that $\ell_{x_1} \le \cdots \le \ell_{x_n}$. Set $V_t = 0$ and, successively for $k = 1, \dots, n$, define*

$$V_{x_k} = \min_{\alpha \in A^{\mathrm{c}}_{x_k}} \max_{b \in B_{x_k}} \frac{c^{\alpha b}_{x_k} + \sum_{\substack{y \in X \setminus \{x_k\} \\ \ell_y < \ell_{x_k}}} p^{\alpha b}_{x_k y} V_y}{1 - p^{\alpha b}_{x_k x_k}}. \tag{21}$$

Then $V = U$.

Proof. By theorem 4.1, every nonself successor under a causal action has strictly smaller ℓ and has therefore already been evaluated. Equation (21) is exactly the scalar fixed-point formula from theorem 3.3 applied to (15). Induction over the topological order yields $V = U$. □

Remark 4.3 (Comparison with the source paper)**.** *Gaspard and Vladimirsky call an SSP explicitly causal when the relevant dependency graph is acyclic and observe that Dijkstra's and Dial's methods are not really needed in that case [2, Section 3.2]. The certificate manuscript becomes explicitly causal after the certified pruning. It therefore does not address the nontrivial situation in which the full local transition structure contains genuine cycles but an unknown optimal policy induces a value-monotone order that must be discovered at run time.*

Remark 4.4 (Complexity)**.** *As written, the proposed algorithm recomputes $D_S(x)$ for every unfinalized state at every iteration. Even if every local evaluation takes constant time, this is $\Theta(n^2)$ work. No event-driven update rule or heap-based complexity proof is provided. In contrast, the direct sweep (21) takes one evaluation per state after sorting by ℓ; its work is $O(n \log n + L)$, where L is the total local input size needed to evaluate all action-response branches.*

5 The certificate is equivalent to the implicit target property

We next show that the certificate does not characterize a stronger, independently checkable structural condition. It can simply encode the unknown value and an unknown optimal selector.

Definition 5.1 (Strictly causal optimal selector)**.** *A selector $\alpha = (\alpha_x)_{x \in N}$ with $\alpha_x \in A_x$ is called δ-causal and optimal if, for every $x \in N$,*

$$\max_{b \in B_x} Q^{\alpha_x b}_x(U) = U_x, \tag{22}$$

and, for every $b \in B_x$ and $y \neq x$,

$$p^{\alpha_x b}_{xy} > 0 \quad \Longrightarrow \quad U_y + \delta \le U_x. \tag{23}$$

Theorem 5.2 (Equivalence of certificate existence and strict causality)**.** *Assume finite action sets and uniform transience (3). The following are equivalent:*

(a) *There exists a certificate $(\ell, u, \delta, (A^{\mathrm{c}}_x)_x)$ satisfying* (C1)–(C4)*.*

(b) *There exists a δ-causal optimal selector in the sense of theorem 5.1, for some $\delta > 0$.*

Proof. Assume (a). By theorem 3.4, for each x one may choose a minimizing action $\alpha_x \in A^{\mathrm{c}}_x$ in (15). It satisfies (22), and *(C2)* together with $\ell \le U \le u$ gives (23). Thus (b) holds.

Conversely, assume (b). Since $c^{ab}_x \ge 0$, monotone iteration from 0 gives $U \ge 0$. Define

$$\ell = u = U, \qquad A^{\mathrm{c}}_x = \{\alpha_x\}. \tag{24}$$

Condition *(C1)* holds with equality because $TU = U$. Uniform transience implies $p^{ab}_{xx} \le \rho < 1$ for every action pair, since $p^{ab}_{xx} w_x \le \sum_y p^{ab}_{xy} w_y \le \rho w_x$. Thus (5) holds, while (6) is exactly (23). Condition *(C4)* is immediate.

It remains to verify *(C3)*. Fix an excluded action $a \neq \alpha_x$, and choose $b_a \in \arg\max_{b \in B_x} Q^{ab}_x(U)$. For every $b' \in B_x$, optimality of α_x gives

$$Q^{\alpha_x b'}_x(U) \le \max_b Q^{\alpha_x b}_x(U) = U_x \le \max_b Q^{ab}_x(U) = Q^{ab_a}_x(U).$$

The box $[\ell, u]$ in (24) is the singleton $\{U\}$, so this is precisely *(C3)*. Hence (a) holds. □

Corollary 5.3 (The certificate can contain the answer)**.** *In the manuscript's finite pure-action model, certificate existence is neither stronger nor more primitive than the desired strict monotone-causality property. A valid certificate may be formed from the exact unknown value function and an exact unknown optimal selector.*

This does not make verification of a supplied certificate logically circular: one can check a proposed $(\ell, u, A^{\mathrm{c}})$ without separately knowing U. It does show that the manuscript has not derived a useful a priori criterion from primitive local data. Without an independent method for constructing a nondegenerate box and causal action sets, the certificate may be as difficult to find as the solution itself. This is exactly the distinction emphasized in the source paper between implicit conditions involving the optimal policy or value and local structural criteria available before solving the dynamic program [2, Section 3.2].

6 Further errors and scope limitations

6.1 Weakly pruned actions may still be optimal

Example 6.1 (A certified "noncausal" action can be optimal)**.** *There is one nonterminal state x, one Max action, and two Min actions α and a. Both actions cost* 1 *and move to t with probability one. Take*

$$\ell_x = u_x = 1, \qquad \delta = 1, \qquad A^{\mathrm{c}}_x = \{\alpha\}.$$

Uniform transience holds with $w_x = 1$ and $\rho = 0$. Conditions (C1)–(C4) *all hold, with* (C3) *holding at equality. Nevertheless, both α and the excluded action a are optimal. Therefore the manuscript's abstract-level claim that noncausal actions are "never optimal" is false. The correct statement is only that they can be removed without changing the value.*

6.2 The operator is not the general simultaneous-game operator

For a standard finite simultaneous zero-sum stochastic game, players may need to randomize at each state. The one-state example below already separates the two models.

Example 6.2 (Pure min–max versus simultaneous game value)**.** *There is one nonterminal state x, all action pairs transition immediately to t, and the cost matrix to Min is*

$$C = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The pure sequential operator (2) *gives*

$$\min_{a \in \{1,2\}} \max_{b \in \{1,2\}} C_{ab} = 1.$$

In the simultaneous zero-sum game, each player mixes uniformly and the value is $1/2$. Thus (2) does not describe the general finite simultaneous SSP game. It describes a sequential robust-control game, or a simultaneous game only when suitable pure saddle-point assumptions are imposed. A general finite simultaneous game can instead be written in min–max form by replacing the displayed pure action sets with the compact simplices of mixed actions and reformulating all certificate conditions uniformly over those simplices; that is not the finite pure-action theorem stated in the manuscript. Standard SSP-game formulations explicitly allow randomized policies or equivalently use compact action spaces of mixed actions [4, 6].

This scope issue does not invalidate theorem 3.4; it limits what that theorem proves relative to the general stochastic-game question.

6.3 Uniform transience forces every policy pair to terminate rapidly

Proposition 6.3 (Strength of uniform transience)**.** *Under (3), for every history-dependent choice of action pairs and every initial state x, the target hitting time τ satisfies*

$$\Pr_x(\tau > k) \leq \frac{w_x}{\min_{y \in N} w_y} \rho^k, \tag{25}$$

and therefore $\mathbb{E}_x[\tau] < \infty$.

Proof. Let X_k be the state at time k and define $W_k = w_{X_k} \mathbf{1}_{\{X_k \neq t\}}$. Conditional on the history and $X_k = x \in N$, whichever action pair is selected, (3) gives $\mathbb{E}[W_{k+1} \mid \mathcal{F}_k] \leq \rho W_k$. Hence $\mathbb{E}_x[W_k] \leq \rho^k w_x$. On $\{\tau > k\}$ one has $W_k \geq \min_{y \in N} w_y$, which yields (25) by Markov's inequality. Summing the tail bound gives finite expected hitting time. $\square$

Thus every policy pair is nonprolonging. This is much stronger than standard SSP-game assumptions, which can permit prolonging policy pairs while still guaranteeing a well-defined value under appropriate cost conditions [4, 6].

6.4 The Dial statement is an implementation assumption, not a causality theorem

The manuscript's bucket discussion assumes that every generated exact label lies on a grid and that newly generated labels remain within a fixed number of buckets of the current minimum. Under those assumptions, a circular bucket queue can of course replace a comparison heap. But neither property is derived from the primitive game data, and no same-bucket independence or complexity theorem is proved. The statement is therefore a data-structure observation, not an analogue of a substantive δ-causality condition of the kind developed in [2].

6.5 The transience feasibility claim is not linear as written

The manuscript states that (3) is a set of linear inequalities in w and ρ. Jointly in those unknowns, the term ρw_x is bilinear. For a fixed candidate ρ, feasibility in w is a linear problem; one may then search over ρ. The joint statement is nevertheless false as written.

7 Final classification

The open direction in [2] concerns useful criteria for label-setting in general stochastic shortest-path games, and the problem entry emphasizes criteria stated a priori in local primitive data [3]. The manuscript establishes a correct verification theorem for the following restricted situation:

> A global value box is supplied; every action outside a chosen subset is robustly prunable throughout that box; and the retained transition supports are already strictly ordered by the supplied lower bounds.

Once these conditions hold, the reduced system is a triangular min–max dynamic program with removable self-loops. The direct sweep in theorem 4.2 solves it. Moreover, by theorem 5.2, existence of the certificate is equivalent to the desired implicit optimal-selector property and can be witnessed by the unknown solution itself.

Question	Assessment
Is the central theorem mathematically correct?	Yes, after inserting the comparison (17)–(18).
Are all claims in the manuscript correct?	No. The "never optimal" and joint-linearity statements are false, and the algorithmic and Dial claims are overstated.
Does the certificate cover general simultaneous SSP games?	No; it treats a pure sequential min–max operator and imposes uniform transience of every action pair.
Does it reveal nontrivial monotone causality?	No. After pruning, *(C2)* already gives a known topological order.
Does it solve the open problem?	No.
Does it make partial progress?	Only weak, restricted-case progress: it gives a valid certificate-verification and triangular-sweep theorem for a reducibly acyclic subclass.

Accordingly, the mathematically accurate classification is:

$$\boxed{\text{not a solution; at most weak partial progress.}}$$

References

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